EVERY TWO-COLORING OF THE PLANE CONTAINS A MONOCHROMATIC DEGENERATE TRIANGLE WITH SIDES 1, 3, AND 4

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Abstract. We prove that every two-coloring of the Euclidean plane contains three collinear points of one color with consecutive distances 1 and 3. Suppose, for a contradiction, that some two-coloring contains no such triple. We consider the triangular lattice $\Lambda = \mathbb{Z}[\omega]$ with $\omega = e^{2\pi i/3}$, together with its rotated copy $q\Lambda$, where $q = \frac{8+5\omega}{7}$. It shares a sublattice with Λ, so the colorings of the two lattices constrain each other. We show that, in one of its three directions, every row of a unit triangular lattice is colored in monochromatic runs of length exactly four, and that consecutive such rows are related by a fixed shift. Then any two points of the plane at distance 8 have the same color, which is impossible under our assumption.

1. Introduction

Let $\mathbb{E}^2$ denote the Euclidean plane. Suppose that we have a coloring of $\mathbb{E}^2$ using two colors. We are interested in three-point configurations that appear monochromatically in every such coloring. We use the term *triple* to refer to any set of three distinct points.

Definition 1.1. The triple $T(a, b, c)$ is a triple of points forming the vertices of a triangle whose sides have lengths a, b, and c. This includes collinear configurations, which correspond to *degenerate* triangles, with one side equal to the sum of the other two.

The notation $T(a)$ is shorthand for $T(a, a, a)$, the vertices of an equilateral triangle of side length a.

Definition 1.2. A coloring *contains* $T(a, b, c)$ exactly when some congruent copy of it is monochromatic.

A foundational contribution to Euclidean Ramsey theory—which studies such questions, as well as more general ones—was made in the 1970s by Erdős, Graham, Montgomery, Rothschild, Spencer, and Straus. In particular, in their 1973 paper [EGM+73], they showed that $T(a)$ can be avoided: it suffices to divide the plane into parallel stripes of width $\frac{a\sqrt{3}}{2}$ and color them alternately. Later, Jelínek, Kynčl, Stolař, and Valla [JKS+09] also found non-trivial colorings avoiding $T(a)$. To this day, these are the only triples for which such a coloring is known.

In 1975, Erdős et al. [EGM+75] identified several families of triples that are contained in every two-coloring of the Euclidean plane. These include configurations $T(a, b, c)$ such that:

- the ratio $\frac{a}{b}$ equals $2\sin\left(\frac{\theta}{2}\right)$ for some $\theta \in \{30°, 36°, 108°, 120°, 150°\}$[1];
- the triangle has an angle of 30° or 150°;
- the triangle has angles $(\alpha, 2\alpha, 180° - 3\alpha)$ for $0 < \alpha < 60°$, $\alpha \neq 45°$;
- the triangle has angles $(180° - \alpha, 180° - 2\alpha, 3\alpha - 180°)$ for $60° < \alpha < 90°$;
- $T(a, b, c) = T(a, 2a, 3a)$, a degenerate triangle;
- a, b, and c satisfy $c^2 = a^2 + 2b^2$, with $b < 2a$;
- a, b, and c satisfy $a^6 - 2a^4b^2 + a^2b^4 - 3a^2b^2c^2 + b^2c^4 = 0$;
- a, b, and c satisfy $a^4c^2 + b^4a^2 + c^4b^2 - 5a^2b^2c^2 = 0$;
- a, b, and c are the base, the leg, and the circumradius of an isosceles triangle.

In the following year, Shader [Sha76] added the triples $T(a, b, c)$ for which

- the triangle is right-angled;
- a, b, and c satisfy $a^2 + c^2 = 4b^2$, where $\frac{\sqrt{7}-1}{2}\, b \leq a \leq \frac{\sqrt{7}+1}{2}\, b$ (Shader's original range for a was narrower and was extended in the author's thesis [Yan25]).

[1] Erdős et al. erroneously mention the angle 72° instead of 36° in their paper.

After these classical results, no new triples of this kind were found for over four decades, until Currier, Moore, and Yip [CMY24] proved that every two-coloring contains $T(1,1,2)$. Their result was an inspiration for this research, in which we prove the analogous statement for $T(1,3,4)$. By scaling, it then holds for every $T(a,3a,4a)$ as well.

Theorem 1.3. *For every $a>0$, every two-coloring of $\mathbb{R}^2$ contains $T(a,3a,4a)$.*

Theorem 1.3 is proved in Section 3. Several lemmas about runs are taken from the author's thesis [Yan25] (Section 2.2), as is the idea of overlapping a lattice with a rotated copy, proposed in its conclusion.

2. Preliminaries

2.1. **Notation and key triples.** The *Eisenstein integers* are the complex numbers $a+b\omega$ with $a,b\in\mathbb{Z}$, where $\omega=e^{\frac{2\pi i}{3}}=\frac{-1+i\sqrt{3}}{2}$ is a primitive cube root of unity. We write $\Lambda=\mathbb{Z}[\omega]=\mathbb{Z}+\mathbb{Z}\omega$ for the set of all of them. Then $\omega^2=-1-\omega$, $\bar{\omega}=\omega^2$, and $|a+b\omega|^2=a^2-ab+b^2$. Viewed as a subset of the plane, Λ is the triangular lattice (Figure 1). For $x\in\mathbb{C}$ write $x\Lambda=\{x\lambda:\lambda\in\Lambda\}$. A *unit triangular lattice* is a set $z_0+u\Lambda$ with $|u|=1$. Its three *directions* are $u,u\omega,u\omega^2$, and its *rows* in a direction are the lines parallel to it.

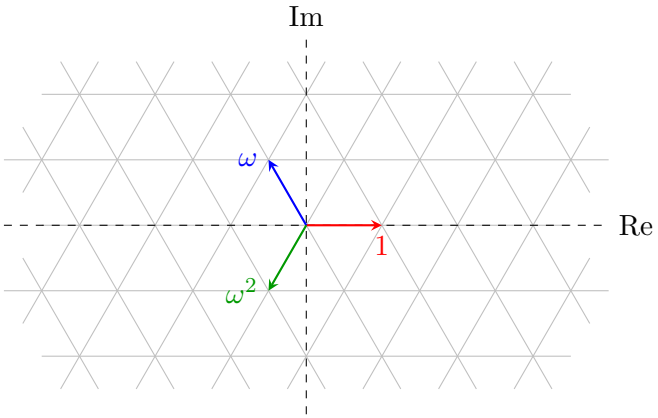


Figure 1. The lattice Λ is spanned by 1 and ω

A two-coloring of the plane is *admissible* if it does not contain $T(1,3,4)$. From now on, we assume that an admissible coloring is given and derive a contradiction. We call the two colors orange and yellow, and write $\chi(p)$ for the color of a point p.

Theorem 2.1 ([EGM+75])**.** *A two-coloring of the plane contains $T(a,b,c)$ if and only if it contains $T(a)$, $T(b)$, or $T(c)$.*

The proof by Erdős et al. [EGM+75] applies equally to degenerate triangles. In particular, for $T(1,3,4)$ it means that a coloring contains $T(1,3,4)$ if and only if it contains $T(1)$, $T(3)$, or $T(4)$.

Definition 2.2. A *key triple* is a congruent copy of $T(1)$, $T(3)$, $T(4)$, or $T(1,3,4)$. In an admissible coloring, no key triple is monochromatic.

2.2. **Runs in a lattice.** We need three facts about a single lattice, all taken from the analysis of $T(1,3,4)$-free colorings in the author's thesis [Yan25, Section 3]. Complete proofs are included here for admissible colorings of the plane. Fix an admissible coloring and a unit triangular lattice L.

Definition 2.3. A *run* is a maximal set of consecutive points of one color on a row.

Lemma 2.4 (runs)**.**

(a) *Every run has at most four points.*
(b) *If a run has at least three points, then the runs next to it on both sides have the other color and at least three points.*

(c) *On each row either every run has at most two points (such a row is called* short*) or every run has three or four points (such a row is called* long*).*

Proof. Identify L with Λ so that the row is $\mathbb{Z}$.

(a) If $0, 1, 2, 3, 4$ have one color, then $\{0, 1, 4\}$ is a monochromatic $T(1, 3, 4)$.

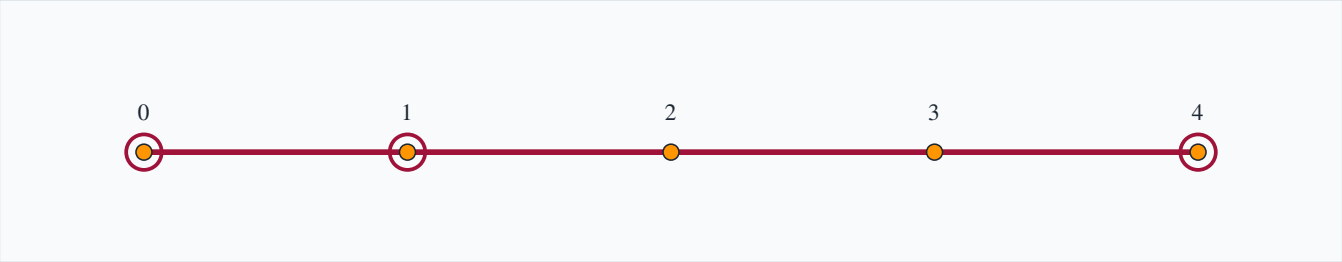


FIGURE 2. Five consecutive points of one color

(b) Let $0, 1, 2$ be the last three points of an orange run, so 3 is yellow. The copy $\{0, 1, 4\}$ of $T(1, 3, 4)$ makes 4 yellow, and the copy $\{1, 2, 5\}$ makes 5 yellow. So the next run is yellow with at least three points. The other side is symmetric.

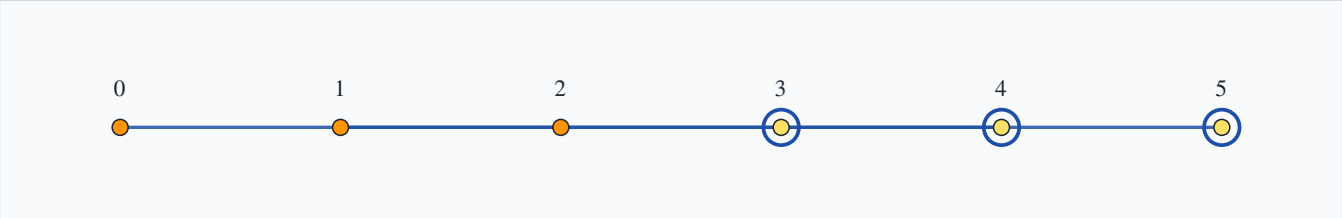


FIGURE 3. The run after an orange run

(c) By (b), if one run on a row has at least three points, so do its neighbors, and by induction all runs on the row. By (a), they have three or four points. □

Lemma 2.5 (adjacent rows)**.** *Two adjacent parallel rows are both long or both short. Hence the rows of one direction are either all long or all short, and we call the direction long or short accordingly.*

Proof. Identify L with Λ so that the rows are $\mathbb{Z}$ and $\mathbb{Z} + \omega$. The reflection $z \mapsto \bar{z} + \omega$ swaps them, so we may assume that $\mathbb{Z}$ is long and $\mathbb{Z} + \omega$ is short. Exchanging colors and translating along the row, let $0, 1, 2$ be the last three points of an orange run of $\mathbb{Z}$. By Lemma 2.4(b), 3 and 4 are yellow. The copy $\{0, 1, 1+\omega\}$ of $T(1)$ makes $1 + \omega$ yellow, and the copy $\{1, 2, 2 + \omega\}$ makes $2 + \omega$ yellow. As $\mathbb{Z} + \omega$ is short, their run is exactly $\{1 + \omega, 2 + \omega\}$, so ω and $3 + \omega$ are orange. The copy $\{3, 4, 4 + \omega\}$ of $T(1)$ makes $4 + \omega$ orange, and ω, $3 + \omega$, $4 + \omega$ form an orange $T(1, 3, 4)$, a contradiction (Figure 4). The second statement follows by induction over the rows of a direction.

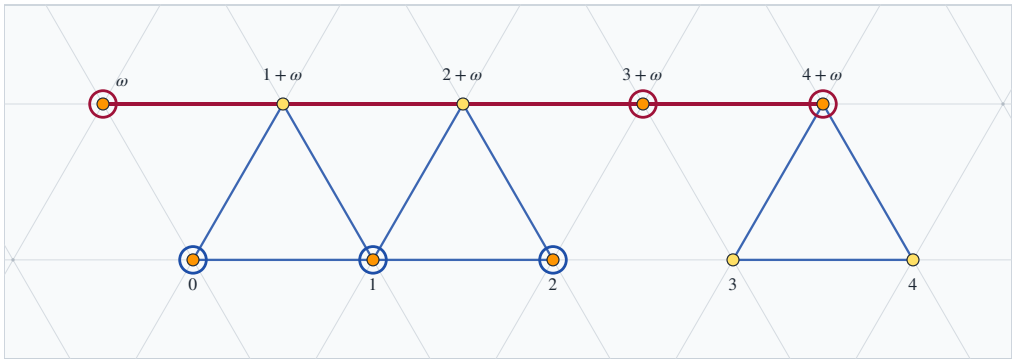


FIGURE 4. A long row below a short row

□

Lemma 2.6 (a long direction)**.** *At least one direction is long.*

Proof. Suppose every row is short, so no three consecutive points of a row have one color. By the pigeonhole principle, two vertices of any unit triangle have the same color. They are adjacent on a row, and after a symmetry of L and exchanging colors we may assume that they are the orange points $0, 1$ of Λ. As rows are short, -1 and 2 are yellow. The copy $\{0, 1, 1+\omega\}$ of $T(1)$ makes $1+\omega$ yellow, and the copy $\{-2, -1, 2\}$ of $T(1, 3, 4)$ makes -2 orange. The copy $\{-1, 2, 2+3\omega\}$ of $T(3)$ makes $2+3\omega$ orange, and the copy $\{-2, 1, 1+3\omega\}$ makes $1+3\omega$ yellow. The points $1+\omega, 1+2\omega, 1+3\omega$ lie on a row, so $1+2\omega$ is orange. Now $1+2\omega$ and $2+3\omega$ are consecutive orange points of the row through ω, $1+2\omega$, $2+3\omega$, and $3+4\omega$, so ω and $3+4\omega$ are yellow. The points -1, ω, and $3+4\omega$ form a yellow $T(1, 3, 4)$, a contradiction (Figure 5).

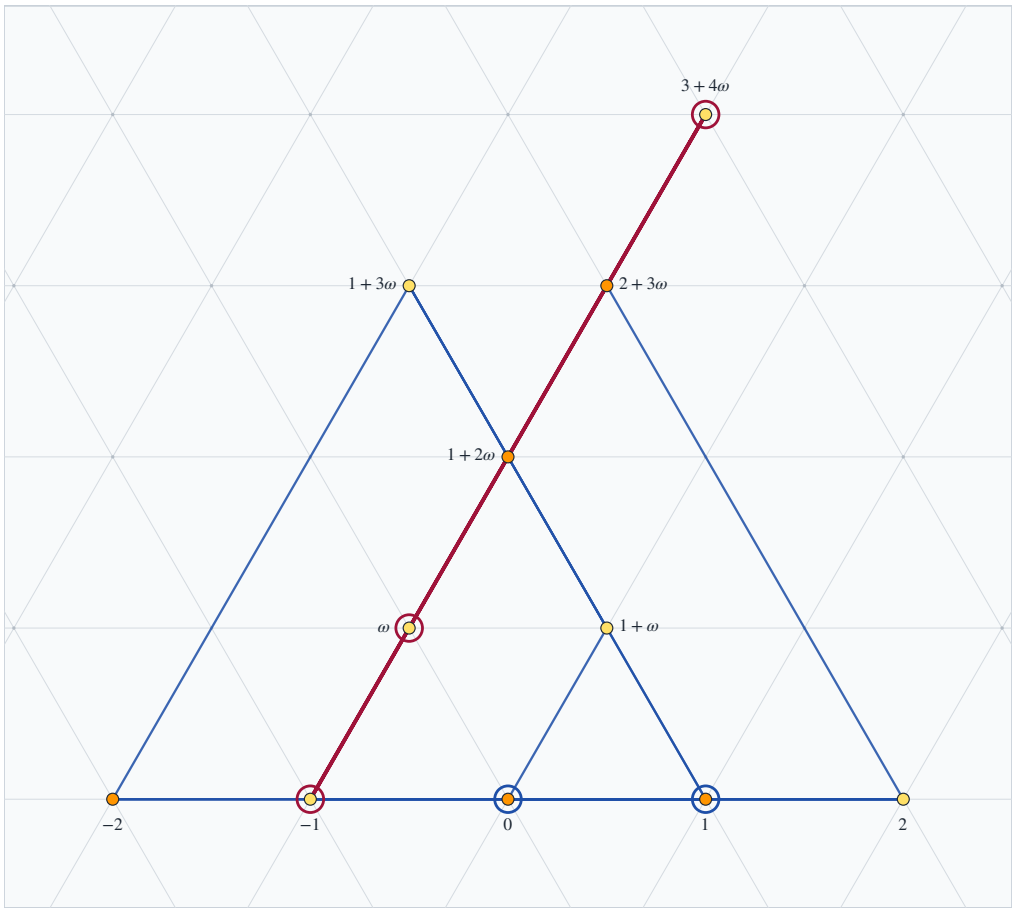


FIGURE 5. If every row is short

□

2.3. **A rotated copy of the lattice.** Section 2.2 alone does not give a contradiction, but it takes away much of the freedom. In an admissible coloring of the plane, every unit triangular lattice has a long direction, and different lattices share points. The idea is therefore to overlap such lattices with many common points (in fact, whole sublattices), so that they constrain each other there.

We want a second unit triangular lattice $q\Lambda$, $|q| = 1$, that is genuinely rotated and shares many points with Λ, since shared points are where constraints pass from one lattice to the other. The points $2-\omega$ and $3+\omega$ of Λ are symmetric about the real axis, so they lie at the same distance from 0. This distance is $\sqrt{7}$, since $3+\omega = \frac{5}{2} + \frac{\sqrt{3}}{2}\,i$ and $\frac{25}{4} + \frac{3}{4} = 7$. Let q be the rotation about 0 that maps $2-\omega$ to $3+\omega$ (Figure 6). It is the multiplication by the quotient of the two points:

$$q = \frac{3+\omega}{2-\omega} = \frac{8+5\omega}{7} = \frac{11+5\sqrt{3}\,i}{14}, \qquad |q| = 1.$$

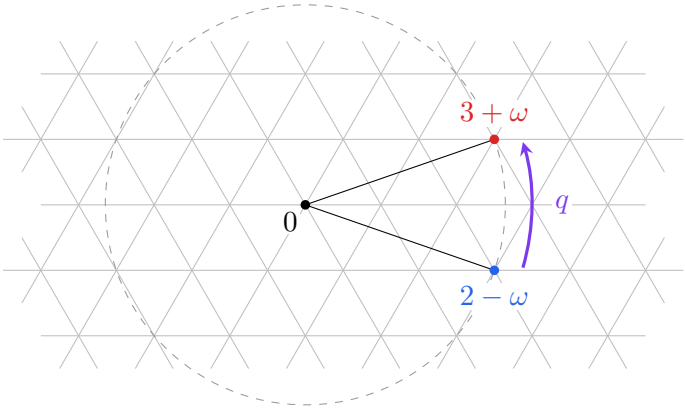


FIGURE 6. The rotation q maps $2-\omega$ to $3+\omega$

Definition 2.7. The rotated lattice is $q\Lambda = \{q(m+n\omega) : m, n \in \mathbb{Z}\}$.

The point $q = q \cdot 1$ lies in $q\Lambda$ but not in Λ, so $q\Lambda \neq \Lambda$. In particular, q is not a sixth root of unity, since all of them lie in Λ. Rotating the multiples of $2-\omega$ gives the multiples of $3+\omega$, as $q\,(2-\omega)\lambda = (3+\omega)\lambda$. Hence the whole sublattice $(3+\omega)\Lambda$ lies in both Λ and $q\Lambda$ (Figure 7). There are no other common points.

Lemma 2.8. $\Lambda \cap q\Lambda = (3+\omega)\Lambda$.

Proof. It remains to show that every point of $\Lambda \cap q\Lambda$ lies in $(3+\omega)\Lambda$. Let $z = q(a+b\omega)$ lie in Λ. As $q = \frac{8+5\omega}{7}$ and $\omega^2 = -1-\omega$,

$$z = \frac{(8a-5b) + (5a+3b)\omega}{7},$$

so 7 divides $8a-5b = 8(a+2b)-21b$, and hence $a+2b$. Let $d = \frac{a+2b}{7}$ and $c = 3d-b$. Then $(2-\omega)(c+d\omega) = (2c+d) + (3d-c)\omega = a+b\omega$, so $z = q(2-\omega)(c+d\omega) = (3+\omega)(c+d\omega)$. □

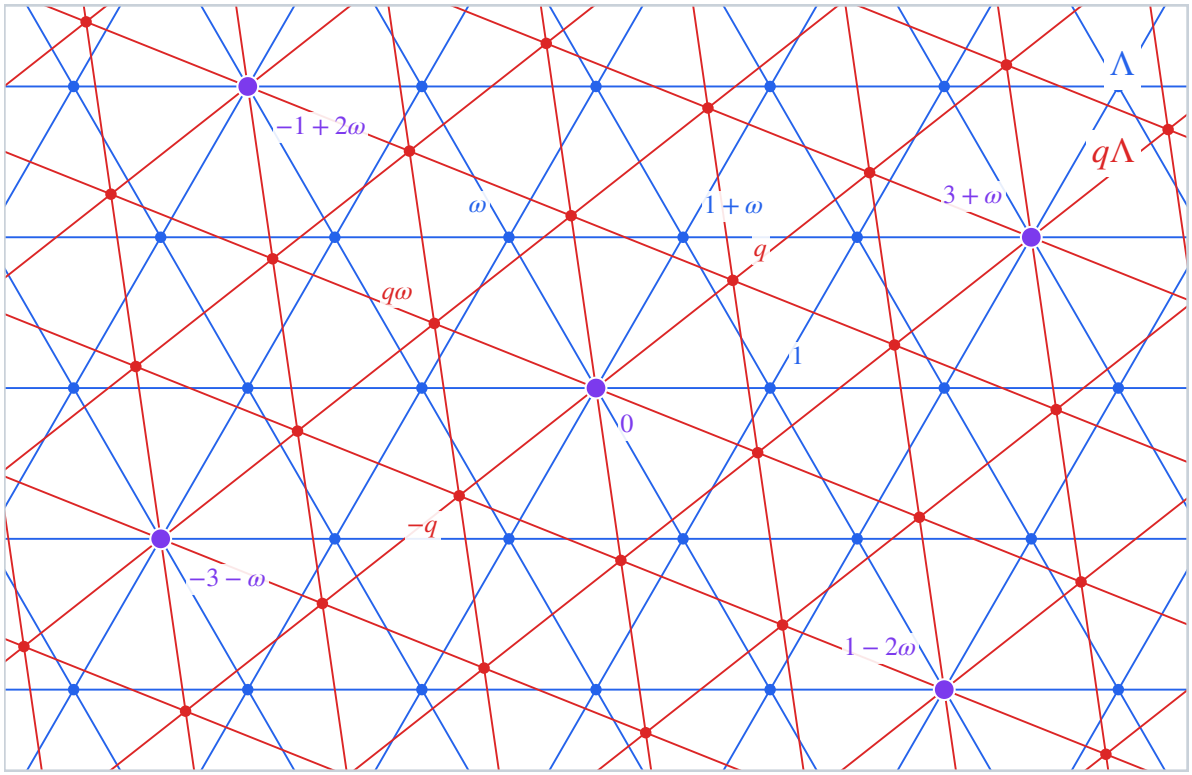


FIGURE 7. Λ (blue), $q\Lambda$ (red), and their common points (purple)

Adding to Λ the rotated copy $q\Lambda$ is already enough to derive Theorem 1.3.

3. Proof of Theorem 1.3

3.1. **Open pairs.** Let L be a unit triangular lattice and u a long direction of L (Lemma 2.6), and put $v = -u\omega^2 = u(1+\omega)$. The three directions of L are u, v, and $v - u = u\omega$. The *rows* of L are its lines in direction u, and the row after R is $R + v$.

Fix a row R. If $z, z+u \in R$ have the same color then the point $z+v$ above them and the point $z+u-v$ below them both have the other color. The remaining points of $R \pm v$, above or below a pair $z, z+u$ with $\chi(z) \neq \chi(z+u)$, are *open*. They come in vertically aligned *open pairs* $\{z+v,\ z+u-v\}$ (Figure 8). For example, in Figure 8 the row is $R = \mathbb{Z} + \omega$, $u = 1$, and $v = 1 + \omega$. The orange points $z = -2 + \omega$ and $z + u = -1 + \omega$ force $z + v = -1 + 2\omega$ and $z + u - v = -2$ to be yellow (blue triangles). The points ω and $1 + \omega$ have different colors, so $1 + 2\omega$ and 0 form an open pair (dashed).

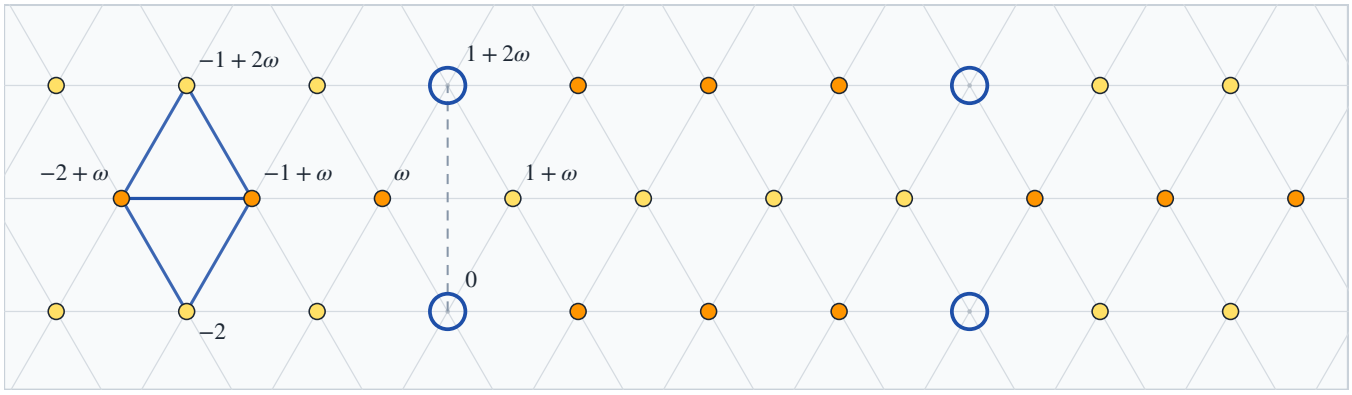


FIGURE 8. Open pairs at the row $\mathbb{Z} + \omega$

Theorem 3.1. *In an admissible coloring of the plane, every open pair has two colors.*

Lemma 3.2 (a monochromatic open pair)**.** *If some open pair is monochromatic then, after an isometry and possibly exchanging colors, $L = \Lambda$, $u = 1$, the open pair is $\{1+\omega,\ -\omega\}$, and*

(a) $-2, -1, 0$ *are orange and* $1, 2, 3$ *are yellow,*
(b) $1+\omega, 2+\omega, 3+\omega$ *and* $-\omega, 1-\omega, 2-\omega$ *are orange,*
(c) $-2+\omega, -1+\omega, \omega$ *and* $-3-\omega, -2-\omega, -1-\omega$ *are yellow.*

Proof. Let $\{z+v,\ z+u-v\}$ be a monochromatic open pair at a row R, so $\chi(z) \neq \chi(z+u)$. Exchanging colors, we may assume that the pair is orange. The reflection in the line through its two points maps L and R to themselves and swaps z with $z+u$, so we may also assume that z is orange and $z+u$ is yellow. Identify L with Λ so that $u = 1$ and $z = 0$. Then $R = \mathbb{Z}$, the point 0 is orange, the point 1 is yellow, and the open pair is $\{1+\omega,\ -\omega\}$ (Figure 9).

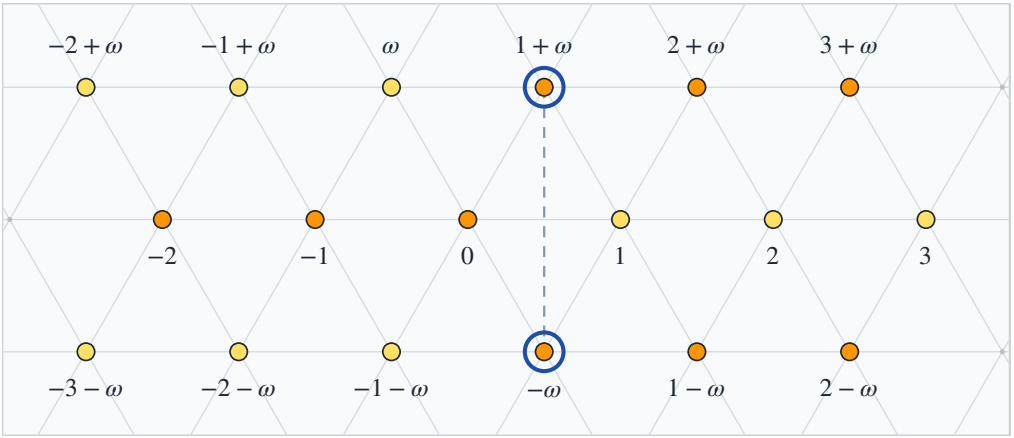


FIGURE 9. A monochromatic open pair

(a) The row $\mathbb{Z}$ is long, so the orange run that ends at 0 contains $-2, -1, 0$. By Lemma 2.4(b), the yellow run that starts at 1 contains $1, 2, 3$.
(b) The points $1+\omega$ and $-\omega$ form the open pair, which is orange. The copy $\{1, 2, 2+\omega\}$ of $T(1)$ makes $2+\omega$ orange, and the copy $\{2, 3, 3+\omega\}$ makes $3+\omega$ orange. Similarly, the copy $\{1, 2, 1-\omega\}$ makes $1-\omega$ orange, and the copy $\{2, 3, 2-\omega\}$ makes $2-\omega$ orange.
(c) The copy $\{-2, -1, -1+\omega\}$ of $T(1)$ makes $-1+\omega$ yellow, and the copy $\{-1, 0, \omega\}$ makes ω yellow. Similarly, the copy $\{-2, -1, -2-\omega\}$ makes $-2-\omega$ yellow, and the copy $\{-1, 0, -1-\omega\}$ makes $-1-\omega$ yellow. By (b) and Lemma 2.4(b), the yellow run that ends at ω contains $-2+\omega, -1+\omega, \omega$, and the yellow run that ends at $-1-\omega$ contains $-3-\omega, -2-\omega, -1-\omega$. □

Lemma 3.3 (two forced points)**.** *Let χ be an admissible coloring of the plane in which the eighteen points of Lemma 3.2 have the colors stated there. Then the points $4+6\omega$ and $-2-3\omega$ are orange.*

Proof. In Λ, the copy $\{-\omega, 1-\omega, 4-\omega\}$ of $T(1,3,4)$ makes $4-\omega$ yellow, and the copy $\{-2+\omega, -1+\omega, -1+2\omega\}$ of $T(1)$ makes $-1+2\omega$ orange.

All other steps take place in the rotated lattice $q\Lambda$. By Lemma 2.8, it contains the multiples of $3+\omega = q(2-\omega)$, and we use six of them:

$$3+\omega = q(2-\omega), \qquad -3-\omega = q(-2+\omega), \qquad -1+2\omega = q(1+3\omega),$$
$$4-\omega = q(1-4\omega), \qquad 4+6\omega = q(6+4\omega), \qquad -2-3\omega = q(-3-2\omega).$$

The last five follow from the first one after multiplying it by -1, ω, $1-\omega$, $2+2\omega$, and $-1-\omega$.

Figures 10 and 11 show these steps on the grid of $q\Lambda$.

Claim. The point q is yellow.

Suppose that q is orange (Figure 10). The copy $\{-3q, 0, q\}$ of $T(1,3,4)$ makes $-3q$ yellow, and the copy $\{-2q, q, -1+2\omega\}$ of $T(3)$ makes $-2q$ yellow. Now $-3q$, $-2q$, and $-3-\omega$ form a yellow $T(1)$, a contradiction. This proves the claim.

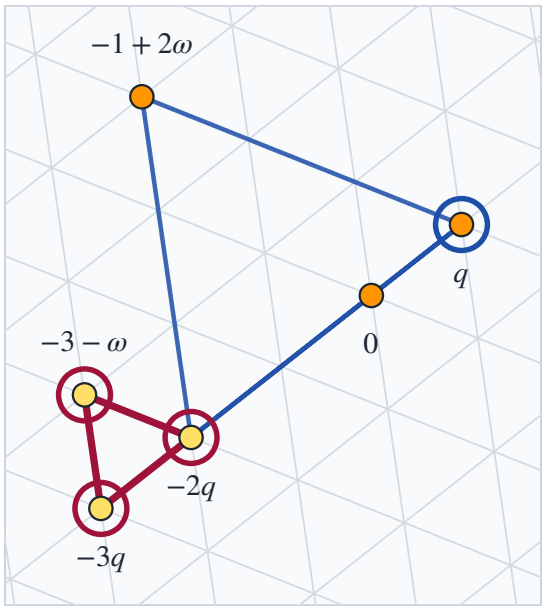


FIGURE 10. An orange q gives a yellow $T(1)$

The remaining steps are shown in Figure 11. The copy $\{q, q(1-\omega), 4-\omega\}$ of $T(1,3,4)$ makes $q(1-\omega)$ orange. Together with $3+\omega$ and $-1+2\omega$, this point forces four yellow points.

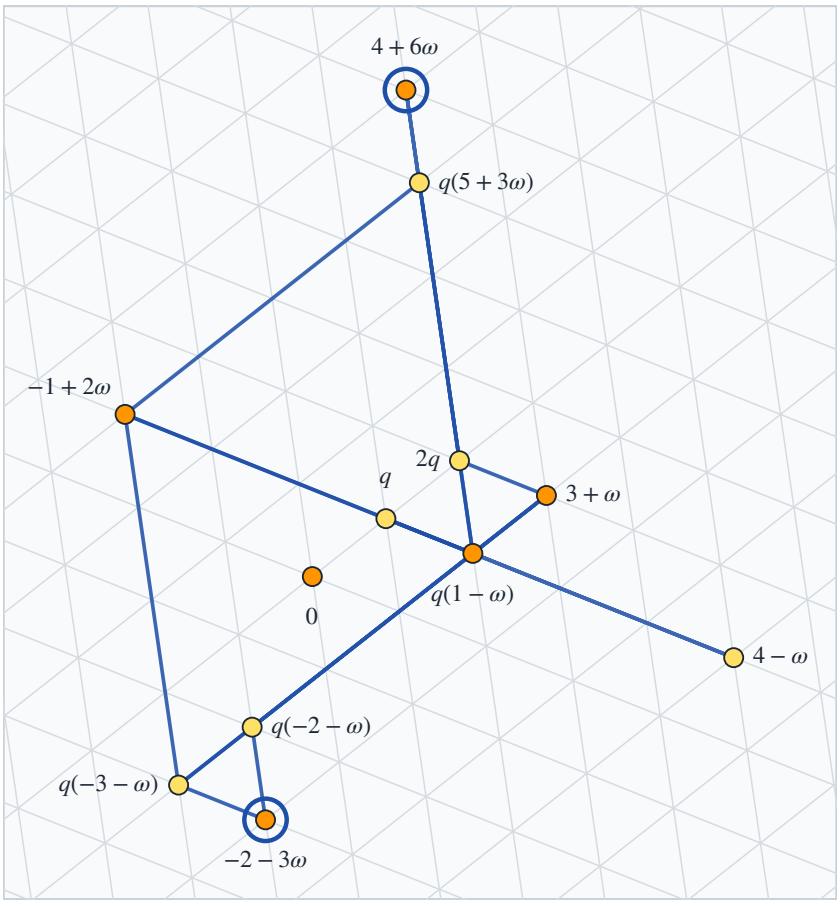


FIGURE 11. The points forced by the orange point $q(1-\omega)$

- The copy $\{q(1-\omega), 3+\omega, 2q\}$ of $T(1)$ makes $2q$ yellow, and the copy $\{q(-2-\omega), q(1-\omega), 3+\omega\}$ of $T(1,3,4)$ makes $q(-2-\omega)$ yellow.
- The points $q(1-\omega)$ and $-1+2\omega$ are at distance 4. The copy $\{q(1-\omega), -1+2\omega, q(5+3\omega)\}$ of $T(4)$ makes $q(5+3\omega)$ yellow, and the copy $\{q(1-\omega), -1+2\omega, q(-3-\omega)\}$ on the other side makes $q(-3-\omega)$ yellow.

Finally, the copy $\{2q, q(5+3\omega), 4+6\omega\}$ of $T(1,3,4)$ makes $4+6\omega$ orange, and the copy $\{q(-3-\omega), q(-2-\omega), -2-3\omega\}$ of $T(1)$ makes $-2-3\omega$ orange. □

Proof of Theorem 3.1. Suppose that some open pair is monochromatic. By Lemma 3.2, we may assume that $L=\Lambda$, $u=1$, and that the eighteen points of the lemma have the colors stated there. By Lemma 3.3, the points $4+6\omega$ and $-2-3\omega$ are orange.

The reflection $z \mapsto \bar{z}$ in the real axis maps the eighteen points with their colors to themselves: it fixes $-2,\dots,3$ and swaps $k+\omega$ with $k-1-\omega$ for $k=-2,\dots,3$. So Lemma 3.3 also applies to the coloring $z \mapsto \chi(\bar{z})$, which is admissible, since the reflection is an isometry. Hence the mirror image $1+3\omega$ of $-2-3\omega$ is orange as well.

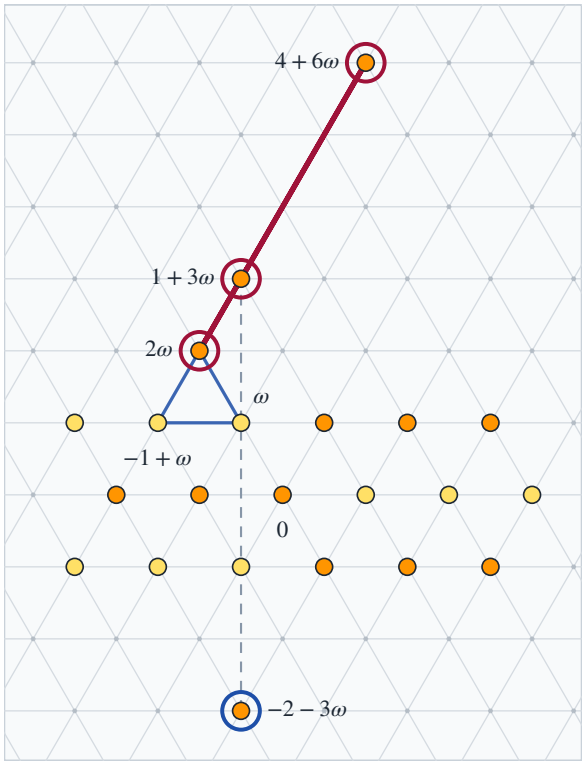


FIGURE 12. The orange copy of $T(1,3,4)$ in Λ

The copy $\{-1+\omega, \omega, 2\omega\}$ of $T(1)$ makes 2ω orange. As $1+3\omega = 2\omega+(1+\omega)$ and $4+6\omega = (1+3\omega)+3(1+\omega)$, the points 2ω, $1+3\omega$, and $4+6\omega$ form an orange $T(1,3,4)$, a contradiction (Figure 12). □

3.2. **Rows.** Every row of L is long, so each of its runs has three or four points (Lemma 2.4(c)).

Lemma 3.4 (fixed shift)**.** *There is $t \in \{v, v-u\}$ such that $\chi(z+t) \neq \chi(z)$ for every $z \in L$.*

Proof. Let $a, a+u, \dots, b$ be a run of a row R with m points, say orange. Then $a-u$ and $b+u$ are yellow. Each of the $m-1$ points $a+v, \dots, b-u+v$ of the row $R+v$ forms a copy of $T(1)$ with two consecutive points of the run, so it is yellow. The same holds for the $m-1$ points $a+u-v, \dots, b-v$ of the row $R-v$. Let A be the run of $R+v$ that contains the first of these two groups of points, and B the run of $R-v$ that contains the second (Figure 13).

On the row $R+v$, the points next to the first group are $a-u+v$ and $b+v$. If $a-u+v$ is yellow, then the copy $\{a-2u+v,\ a-u+v,\ a-u\}$ of $T(1)$ makes $a-2u+v$ orange. If $b+v$ is yellow, then the copy

$\{b+v,\, b+u+v,\, b+u\}$ of $T(1)$ makes $b+u+v$ orange. So A consists of the first group and the yellow points among $a-u+v$ and $b+v$. In the same way, B consists of the second group and the yellow points among $a-v$ and $b+u-v$.

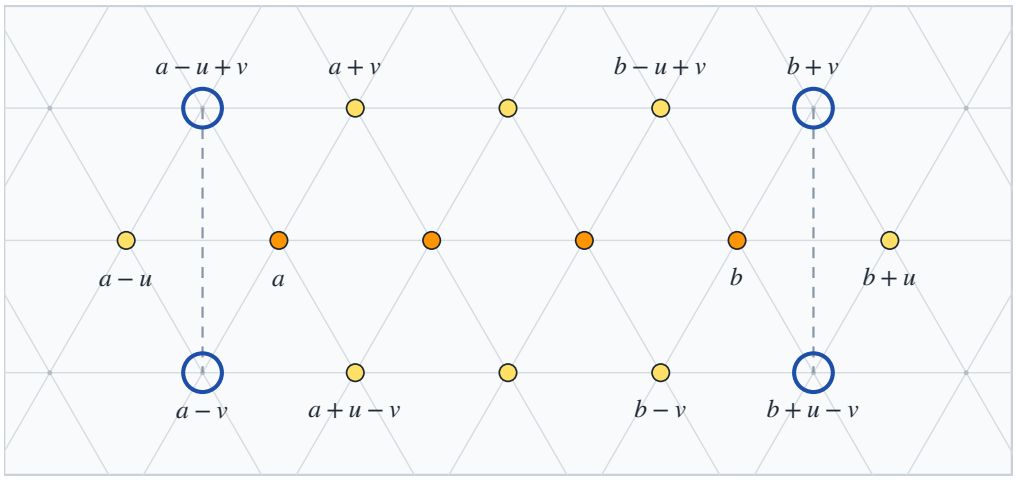


FIGURE 13. A run with its two open pairs

The pairs $\{a-u+v,\, a-v\}$ and $\{b+v,\, b+u-v\}$ are open pairs at the row R. By Theorem 3.1, each of them contains exactly one yellow point, so A and B have $2(m-1)+2=2m$ points together. As m is 3 or 4 and both A and B have three or four points, each of them has exactly m points. Hence exactly one of the points $a-u+v$ and $b+v$ is yellow.

- If $b+v$ is yellow, then $a-u+v$ is orange. By the open pairs, $b+u-v$ is orange and $a-v$ is yellow. So $A=\{a+v,\dots,b+v\}$ and $B=\{a-v,\dots,b-v\}$, and both $\chi(z+v)\neq\chi(z)$ and $\chi(z-v)\neq\chi(z)$ hold for every point z of the run. We say that the run has *shift* v.
- If $a-u+v$ is yellow, then $b+v$ is orange. By the open pairs, $a-v$ is orange and $b+u-v$ is yellow. So $A=\{a-u+v,\dots,b-u+v\}$ and $B=\{a+u-v,\dots,b+u-v\}$, and both $\chi(z+v-u)\neq\chi(z)$ and $\chi(z-v+u)\neq\chi(z)$ hold for every point z of the run. We say that the run has shift $v-u$.

The run after $a,\dots,b$ on R is yellow and starts at $a'=b+u$. The same two cases, with the colors exchanged, show that it has shift v exactly when $a'-u+v$ is yellow. Since $a'-u+v=b+v$, this is exactly when the run $a,\dots,b$ has shift v. So consecutive runs have the same shift, and all runs of R have the same shift t_R. Hence $\chi(z+t_R)\neq\chi(z)$ and $\chi(z-t_R)\neq\chi(z)$ for every $z\in R$.

Finally, let $R'=R+v$ be the row after R, and suppose that $t_{R'}\neq t_R$. Let $z\in R$. Then $\chi(z+t_R)\neq\chi(z)$, and also $\chi(z+t_{R'})\neq\chi(z)$, by the second relation for the point $z+t_{R'}$ of R'. As $\{t_R,t_{R'}\}=\{v,v-u\}$, the neighbors $z+v-u$ and $z+v$ on R' have the same color. This holds for every $z\in R$, so R' has only one color, which contradicts Lemma 2.4(a). Hence $t_{R'}=t_R$, and all rows have the same shift t. □

Lemma 3.5 (four apart)**.** *For every $z\in L$ we have $\chi(z+4u)\neq\chi(z)$ and $\chi(z+8u)=\chi(z)$.*

Proof. Let t be as in Lemma 3.4, so $\chi(x+4t)=\chi(x)$ for every $x\in L$. The points z, $z+4u$, and $z+4v$ form a copy of $T(4)$. If $t=v$, then $\chi(z+4v)=\chi(z)$. If $t=v-u$, then $z+4v=(z+4u)+4t$, so $\chi(z+4v)=\chi(z+4u)$. In both cases, $\chi(z+4u)=\chi(z)$ would make this copy monochromatic. Hence $\chi(z+4u)\neq\chi(z)$, and applying this twice gives $\chi(z+8u)=\chi(z)$. □

In particular, no run of a row has exactly three points, since the points before and after it would be four apart and have the same color. So each row is colored with four orange points, four yellow points, and so on, repeating.

3.3. Equal colors at distance eight.

Proposition 3.6. *In an admissible coloring of the plane, any two points at distance 8 have the same color.*

Proof. Let $|p-p'|=8$ and take a unit triangular lattice $L \ni p$ in which $p'-p=8w$ for a direction w of L. Let u be a long direction of L and $v=u(1+\omega)$, as before. By Lemma 3.4, there is $t \in \{v, v-u\}$ with $\chi(z+t) \neq \chi(z)$ for every $z \in L$, and hence $\chi(z+8t)=\chi(z)$. By Lemma 3.5, $\chi(z+8u)=\chi(z)$. The third direction is $t-u$ or $t+u$, and $\chi(z+8t\pm 8u)=\chi(z)$ as well. So $\chi(p')=\chi(p)$ whichever direction w is. □

Theorem 1.3. *For every $a>0$, every two-coloring of $\mathbb{R}^2$ contains $T(a,3a,4a)$.*

Proof. By scaling, we may assume $a=1$. Suppose χ is admissible, and let A, B, C be collinear with $|A-B|=1$, $|B-C|=3$, and $|A-C|=4$. On one side of the line, the circles of radius 8 about A and B meet in a point P, and those about B and C in a point Q (Figure 14). Proposition 3.6 gives $\chi(A)=\chi(P)=\chi(B)=\chi(Q)=\chi(C)$, so $\{A,B,C\}$ is a monochromatic $T(1,3,4)$, a contradiction. □

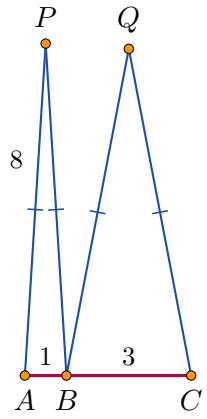


FIGURE 14. The end of the proof of Theorem 1.3

4. Use of AI

The exploration purely on the triangular grid, including the neighborhood argument, was done while writing my thesis [Yan25]. During that work, the idea of overlapping triangular lattices came to my mind as a potential direction, which the thesis mentions in its conclusion: I was driven by the intuition that the structure theorem is far too limiting, that several lattices have to be combined, and that the constraints would then accumulate quickly enough to force a contradiction. GPT-6 Astra discovered that after the overlap the coloring is 8-periodic, and found two statements as a way to split the proof of this periodicity: that no row has a run of exactly three points, and that every open pair has two colors (Theorem 3.1). It also calculated the coordinates of the points needed and derived the proofs. Claude Fable 5.2 found a shorter argument. It proved that every open pair has two colors before the lengths of the runs are known, so that the statement on runs of exactly three became a consequence of Lemma 3.5, and it found the short proof of Theorem 3.1 in Λ and $q\Lambda$, using the reflection $z \mapsto \bar{z}$.

Acknowledgements

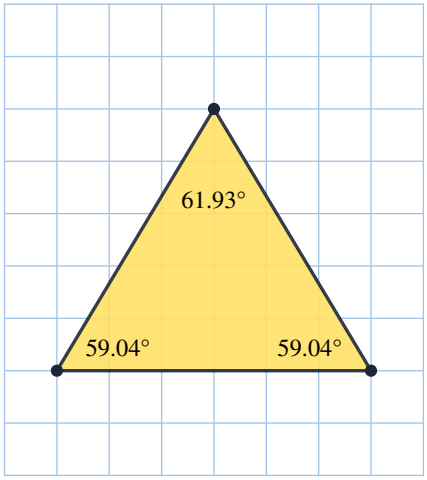
I am grateful to my thesis supervisor Jan Kynčl, whose suggestion to look at this problem created the spark for this work, and for his guidance during my thesis work, which made this paper possible.

Dedication

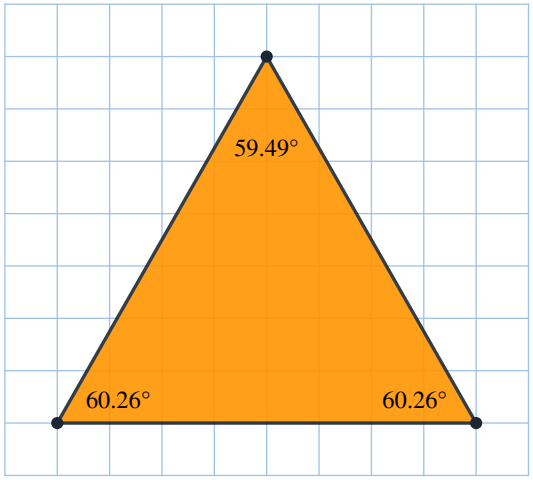
This work is dedicated to the memory of the extraordinary Kharkiv teacher, computer scientist, mathematician, and wartime volunteer Yulia Zdanovska[1] (*04.05.2000–†03.03.2022), who opened my eyes to the wonder of combinatorics, graphs, and algorithms. She spent countless hours patiently teaching me to solve problems in these fields when we were preparing for mathematics and informatics olympiads together, back in the carefree days of our childhood.

I remember us reading a paper on the chromatic number of space while sitting on the Giant Bench in Holosiiv Park in Kyiv, two and a half meters above the ground. It was a sunny, calm day in the forest by the lake full of ducks, and we were eating improvised sandwiches. That is when we discovered that we drew equilateral triangles on grid paper in different ways: I used isosceles triangles with base 6 and height 5, while Yulia used ones with base 8 and height 7. We made a bet on a chocolate bar over whose approximation was better, and calculated the angles. Mine came out as $(59.04°,\ 59.04°,\ 61.93°)$, hers as $(60.26°,\ 60.26°,\ 59.49°)$. She won the chocolate bar.

I wish there were more such afternoons, and more bets for her to win.



base 6, height 5



base 8, height 7

I am eternally grateful for her inspiration and feel honored to have known her so closely.

[1]For more about her life and legacy, see `https://teachforukraine.org/en/t4u-news/ne-hochu-v-garvard-hochu-buty-ministerkoyu-osvity-istoriya-zagybloyi-matematykyni-yuliyi-zdanovskoyi/`.

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